<u>Don't Hate My Spectrophotometer Because It's Janky</u>

Ethan S

Abstract

The visible-light spectrophotometer, long the prerogative of moderately well-funded research and education labs, need remain so no longer! Presented is a description of how a functional spectrophotometer was built over three months, prioritizing parts of old instruments and inexpensive materials found around the lab. The electronic parts that needed to be bought new totaled under a hundred dollars. The resulting instrument proved capable of distinguishing samples of different colors and performing a simple quantitative chemical analysis. A full description of the design and construction is provided in order to enable others to create their own janky spectrophotometers with the parts at hand.

1 – Procedure

This experiment required design refinement and problem-solving in three distinct areas; first to build a functional instrument required an interweaving of hardware and software: the body and the brain, so to speak, respectively, of the spectrophotometer. Finally, the instrument being completed, it had to be put through its paces with simple tests which served to expose limitations and mistakes and inform further refinements to both hardware and software.

1.1 – Hardware

This was the first part of the project begun and seemed deceptively simple. Determining the appropriate interplay of optical components and determining the appropriate wiring proved to be the most complex part of the entire instrument.

1.1.1 – Housing

The size of the spectrophotometer was dictated by the focal length of the lens used (approx. 60 cm). A cabinet with a length this order of magnitude (75 cm) was selected and procured to house the equipment. The interior was coated in a black heat-resistant spray enamel, which served a dual purpose: primarily to reduce light reflection within the box, but secondarily to protect the wooden cabinet from the heat of the bulb. Some interior components received a coat of black spray paint as well. Black foam that had lined an instrument case of some sort was repurposed to keep unwanted stray light out of the cabinet: it was cut into long, inch-thick[1] strips which were glued around the cabinet door such that when it was closed, all gaps were filled. Conveniently, this also facilitated the ingress of electrical power cords into the box, as they could simply be routed through a seam between strips, creating no additional holes needing to be filled.

[1] That is, 2.5-centimeters-thick

1.1.2 – Light Source

The bulb used was a tungsten-halogen capsule repurposed from an old spectrophotometer.[2] It was rated for 12 V and 50 W, meaning its hot current requirement was

$$\frac{50\ \text{W}}{12\ \text{V}} \approx 4.2\ \text{A}$$

When cold, however, the bulb's tungsten wire has nearly no resistance, meaning it briefly draws a very high current. To allow for this without tripping the automatic overload protection on the power supply (an adapter transforming standard 120 VAC at 60 Hz from a wall outlet into 12 V at up to 6 A),[3] a thermistor with suitable qualifications (5 Ω 20 A)[4] was wired in series with the bulb. The thermistor is the inverse of a lightbulb in that its resistance is high when cold but low when hot, as opposed to the bulb, which has low resistance when cold, but high when hot. This tradeoff was exploited to keep the power adapter's safety setting from engaging. At any time one or the other of the items is providing sufficient resistance to keep the current within the adapter's rated range. To keep the light contained and focused, the bulb was mounted inside a metal housing recovered from an old spectrophotometer. This had an open base, but otherwise enclosed the bulb save for a small window in one side. A metal slit-containing strip likewise salvaged from an old instrument was taped over this window.

1.1.3 – Light Path

Two mirrors and a lens further manage the light's path as it traverses the spectrophotometer. These were all taken from overhead projectors and placed in the corners of the spectrophotometer, the

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[2] OSRAM Sylvania 64604

[3] Kastar AC Adapter

[4] Vishay Amatherm AS325R020

first mirror on its own, but the second inside of its original casing with slots for two large lenses. The first of these was removed, but the second was left in. As mentioned above, the light source is at the focal point of the lens; this allows the lens to collimate the divergent rays, making them parallel when they fall on the diffraction grating. The diffraction grating used (from an old spectrophotometer) is specifically a reflection grating: rather than acting as a lens, it acts like a mirror, except that it reflects different wavelengths of light at different angles. When the incident light rays are parallel, it creates a well-defined spectrum.

1.1.4 – Wavelength Control

The diffraction grating was mounted onto a stepper motor, which enabled its position to be reliably controlled by a Raspberrry Pi computer. The motor is a six-wire bipolar stepper repurposed from an old instrument, but the driver[5] used to interface with the Pi is designed for a four-wire bipolar stepper motor. This means that the two wires on the motor which connected to the center taps of its two coils had to isolated, effectively turning the motor into a four-wire motor. However, since the motor was being repurposed, no schematic was on hand to determine which these two wires were, or how the other four wires comprised the ends of the two coils. A multimeter was used to determine which wires were ends of the same coil: the measured resistance between two ends of the same coil was .57 Ω, while the resistance between one end and one center tap was roughly half that at .29 Ω, and the resistance between two wires connected to different coils was of course infinite (no electrical connection). That being determined, it remained to use trial and error to discover how to attach these two pairs of wires to the A+, A-, B+, and B- terminals of the driver.

[5] Pololu TB67S279FTG

A second highly important step to using the motor driver was setting the trimmable potentiometer so that the current flowing through the coils maintained a safe limit. With this particular driver, the current in amps is equal to VREF (reference voltage) in volts times .556, or

$$\mathrm{VREF} = \frac{\text{Current}}{.556\ ^{\mathrm{A}}/_{\mathrm{V}}}$$

The motor’s current rating (which was fortunately printed on the bottom) being .4 A means that VREF had to be set no higher than

$$\mathrm{VREF} = \frac{.4\ \mathrm{A}}{.556\ ^{\mathrm{A}}/_{\mathrm{V}}} = .719\ \mathrm{V}$$

to ensure the motor’s safety. To set VREF, the motor’s logic power source (i.e. the Raspberry Pi) was turned on, the DC voltage between the logic ground pin and the VREF test point (a small exposed circle on top of the driver board) was measured, and the trimmable potentiometer was adjusted with a screwdriver until the multimeter showed the desired voltage reading. The driver is capable of inducing microsteps up to $^1/_{32}$; however, this was deemed unnecessarily precise given the limitations of the monochromator slit’s width, and $^1/_8$ was selected as the microstep resolution. Since the motor turns

$$\frac{1\ \cancel{\text{revolution}}}{200\ \text{steps}} \times \frac{360°}{\cancel{\text{revolution}}} = \frac{1.8°}{\text{step}}$$

the diffraction grating moves

$$\frac{1\ \cancel{\text{step}}}{8\ \text{microsteps}} \times \frac{1.8°}{\cancel{\text{step}}} = \frac{.225°}{\text{microstep}}$$

The spectrum from the diffraction grating was able to register as higher than the background noise on about 105 microsteps, meaning that the detectable spectrum covers an angle of roughly

$$105\ \text{microsteps} \times \frac{.225°}{\text{microstep}} \approx 24°$$

Note that this does not mean the spectrophotometer is able to distinguish 105 different wavelengths; see below.

1.1.5 – Sample Holding

A cuvette holder salvaged from a retired spectrophotometer was glued inside one half of one half of a small box. The best way at present to exchange cuvettes unfortunately requires that the entire quarter box be removed from the cabinet, the old cuvette taken out, the new one put in its place, and the entire quarter box replaced. Fortunately, half a hinge sticking off the side of the side of the quarter-box proved exactly the right size to be useful for aligning the cuvette in front of the photodiode: as long as the hinge is touching the inside of the cabinet, the cuvette is in the right place. Although this minor variability does not appear to have had a significant effect on readings, no numerical analysis has yet been performed. The box had a hole for the cuvette drilled in the top, and a slit cut in front of where the cuvette may set. The light reaching the photodiode was too little to produce a strong signal when filtered through a truly or nearly monochromating slit, so the entrance to the sample chamber was left significantly wider (about 4 mm). Although this reduces wavelength specificity, it does not render spectra completely devoid of features, so the tradeoff was judged worthwhile.

1.1.6 – Light Sensing

The light passing through the sample (or blank) was measured by a combination photodiode and transimpedance amplifier chip.[6] This outputted an analog voltage proportional to the intensity of the light falling on it. Because the Pi Zero cannot measure an analog voltage, this signal had to be

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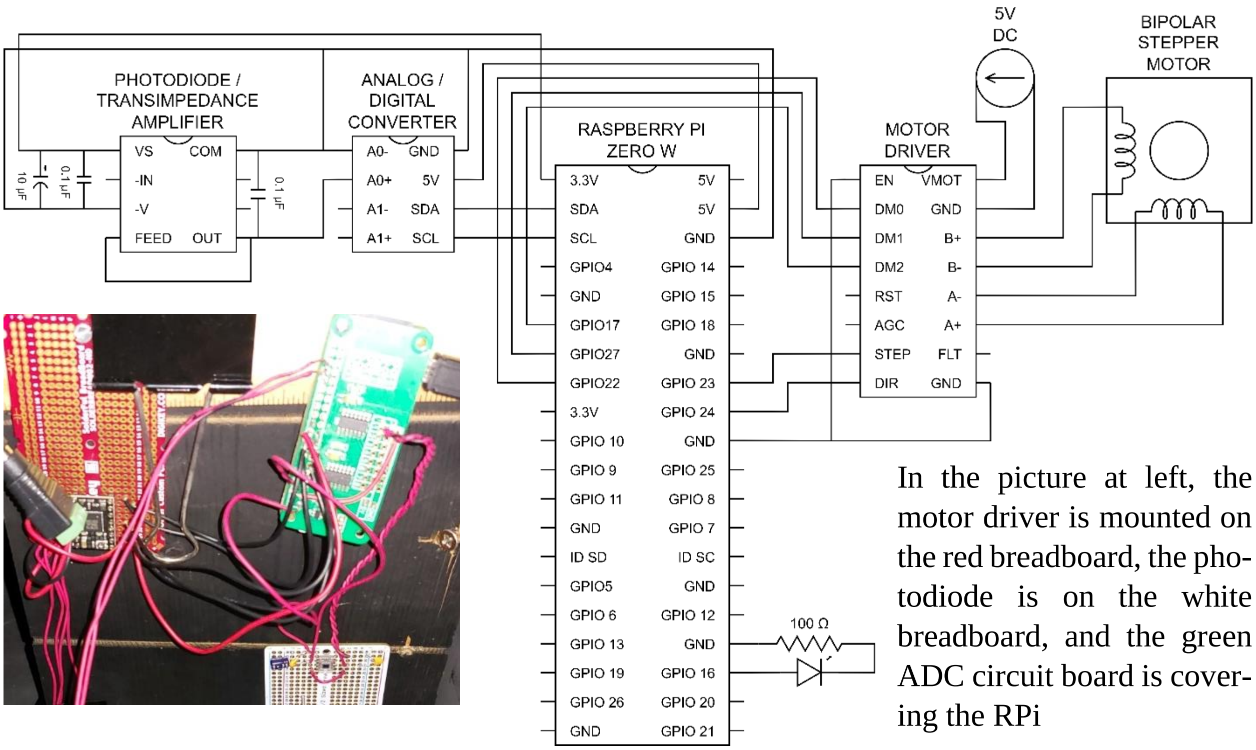
[6] Texas Instruments OPT101

transformed by an analog digital converter[7] into an 18-bit digital signal. Although the ADC used could convert into 16-bit values at a much higher speed, this necessarily gave only one fourth (2^{-2}) the precision, which was unacceptable given the extremely low light levels in question.

1.1.7 – Wiring



In the picture at left, the motor driver is mounted on the red breadboard, the photodiode is on the white breadboard, and the green ADC circuit board is covering the RPi

The wiring diagram leaves little needing to be said. A few points may be clarified, however:

- The Pi has its own power source (a microUSB cable not shown on the schematic)
- The Common and Output wires from the photodiode to the ADC ought to be left as short as possible and twisted tightly together to prevent external interference with the signal
- All the connections from the right side of the ADC to the Pi were automatically taken care of by seating the ADC as a "hat" atop the Pi's GPIO pins

1.2 – Software

[7] ABElectronics ADC Differential Pi

Although many preliminary Python scripts were created to test the building blocks of the spectrophotometer's system, ultimately only three programs were needed to create a fully-functioning instrument. The first two, though based on scripts written by ChatGPT were heavily altered subsequently. The third was produced by Google Gemini very near its final form. All of these scripts may be downloaded from GitHub.

1.2.1 – spectrophotometer.py

This python script is run on the Raspberry Pi itself. It simultaneously monitors for a start trigger, executes that trigger to scan a blank (saving the data as a referenceable list) or a sample (writing a .csv file featuring an ordinal index, a timestamp, the reading, the corresponding blank, and the calculated transmittance and absorbance), steps the motor, and reads the photodiode's output. When it has finished all that, it sends the .csv file via SCP (Secure CoPy) to a specific directory on the host computer. Absorbance calculation deserves a word of explanation. Absorbance is calculated as

$$A = -\log(\frac{P_s}{P_o})$$

where P_s is the power (intensity) of the light having passed through the sample, and P_0 is the power of the light at the same wavelength having passed through a blank.[8] Given the far from complete light insulation within the box (i.e. shielding of the photodiode from the light of the bulb was imperfect), there was a significant background light reading observed even when no

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[8] https://chem.libretexts.org/Bookshelves/Analytical_Chemistry/Instrumental_Analysis_(LibreTexts)/13%3A_Introduction_to_Ultraviolet_Visible_Absorption_Spectrometry/13.02%3A_Beer's_Law

light was being transmitted through the sample. This value (.007 V) was therefore subtracted from all light readings, making the absorbance calculation

$$A = -\log(\frac{P_s - P_b}{P_o - P_b})$$

where P_b is the power of the background light. This provided much sharper spectra with more reasonable absorbance values.

1.2.2 – spectrophotometer_control.py

This is the corresponding script that runs on the controlling computer. It creates a graphical input window that allows the user to select running a blank or a sample, and to input a human-readable name for a sample spectrum (to which will ultimately be appended a timestamp). It sends this information to the Pi, and then monitors the spectrum directory. When it sees that a new file has been written to this directory, it automatically opens the .csv spreadsheet using Tkinter and plots absorbance vs. index value for an immediate glimpse of the output.

1.2.3 – spectrum_compare.py

This third script also runs on the controlling computer, and is capable of plotting multiple spectra on the same graph. This makes it easy to compare and contrast two different samples, or different scans of the same sample.

1.2.4 – Other Python Scripts

Two other scripts, though not strictly necessary to a functioning spectrophotometer, manage in very few lines to only a few lines each to serve very useful purposes: one program that is set to run on startup turns on an LED light, serving as an indicator that the Pi is ready for use. The other one simply reads the ADC value repeatedly as quickly as possible and offers a visual output (useful to check alignment).

1.3 – Tests

Having created this instrument, there remained to test its functionality. Given that due to time and budget constraints it has a limited intended range of use, these tests were simple: one qualitative and one quantitative.

1.3.1 – Color Differentiation

Color standards were prepared by diluting one drop of food coloring (red, yellow, green, and blue) in 20.0 mL deionized water. Three spectra were taken of each of these solutions to determine if the spectrophotometer could differentiate by color, and if its measurements were reproducible.

1.3.2 – Quantitation

The blue solution was subsequently subjected to a series of dilutions to create a standard curve capable of determining the concentration of an unknown. First 15 mL of the initial solution (100% concentration) was diluted to a volume of 25 mL, the difference being made up by deionized water. The concentration of the new solution was (by rearranging $C_1V_1 = C_2V_2$—which simply means that the product of concentration and volume is constant—to get $\frac{C_1V_1}{V_2} = C_2$):

$$\frac{100\% \cdot 15\text{ mL}}{25\text{ mL}} = 60\%$$

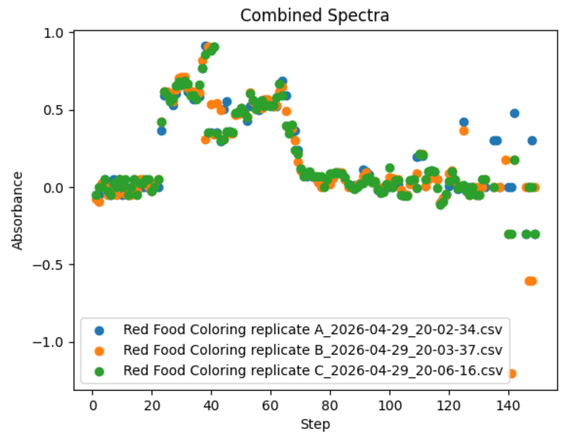
20 mL of this solution was likewise diluted to 50 mL, creating a solution of strength

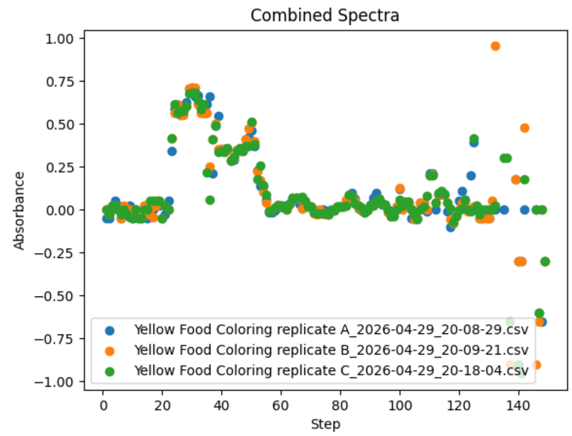
$$\frac{60\% \cdot 20\text{ mL}}{50\text{ mL}} = 24\%$$

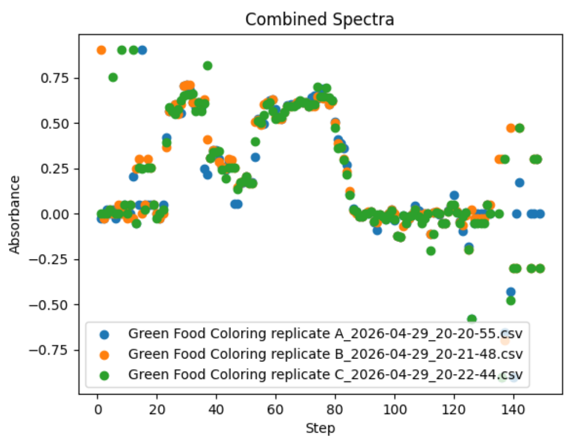
25 mL of this solution was diluted to 50 mL, creating a solution with a relative concentration of

$$\frac{24\% \cdot 25\text{ mL}}{50\text{ mL}} = 12\%$$

An "unknown" solution was created by diluting one of these four solutions by a different amount, but its exact provenance was left undisclosed so that its concentration could be determined by spectroscopic analysis without prejudice.







Replicate spectra of different colors of food dye

Top left: Red; top right: Yellow; bottom left: Green

Absorbance is in arbitrary units

Increasing step number corresponds to lower frequency (longer wavelength) radiation; i.e. 0 is ultraviolet, 150 is infrared

2 – Results and Data

2.1 – Spectra (Color differentiation)

Notice that although there is some variation between different spectra of the same sample, in each graph, all three spectra follow the same basic contours between indices 20 and 100 (the approximate ends of the visible spectrum).[9] Indices less than 20 or greater than 100 are subject to wildly

[9] Note that it has not at this time proved feasible to calibrate the spectrophotometer’s readings to actual wavelengths; the index of each reading (the number of microsteps the motor takes to attain that point) is given instead. In all the spectra presented, a low index corresponds to a higher-

fluctuating calculated absorbances, because past these bounds almost no light from the diffraction grating is reaching the photodiode, and any difference between the measured value and the reference value from the blank for the same index is random fluctuation due to light reaching the photodiode from around the edges of the sample holder, rather than passing through the sample.

The differences between two spectra derived from samples of different colors are far greater than the differences between any two spectra derived from the same sample. This supports that the goal of a spectrophotometer capable of taking simple spectra and differentiating samples based on overall absorbance patterns was met.

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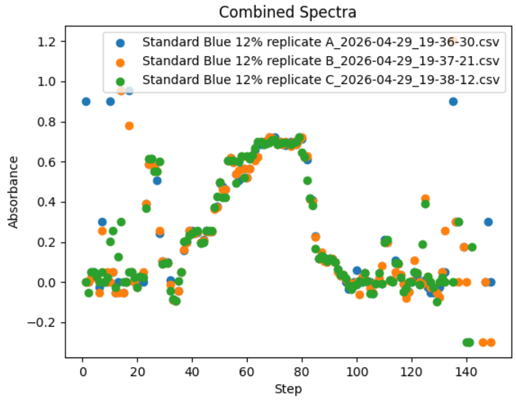
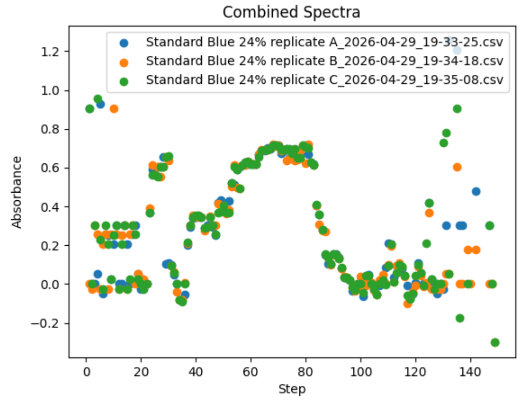
frequency, lower wavelength light signal, and vice versa: a high index corresponds to a lower-frequency, higher-wavelength signal.


Combined Spectra
Standard Blue 12% replicate A_2026-04-29_19-36-30.csv
Standard Blue 12% replicate B_2026-04-29_19-37-21.csv
Standard Blue 12% replicate C_2026-04-29_19-38-12.csv
Absorbance
Step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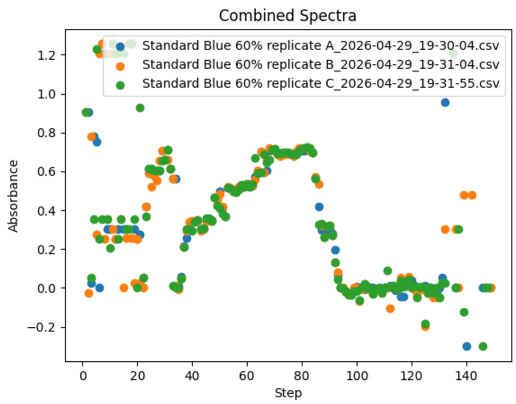
Standard Blue 24% replicate C_2026-04-29_19-35-08.csv
Absorbance
Step


Combined Spectra
Standard Blue 60% replicate A_2026-04-29_19-30-04.csv
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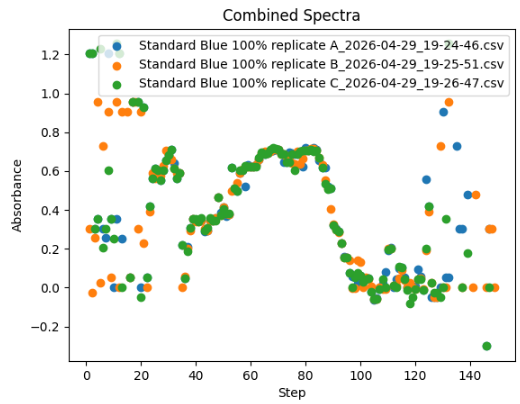
Standard Blue 60% replicate C_2026-04-29_19-31-55.csv
Absorbance
Step


Combined Spectra
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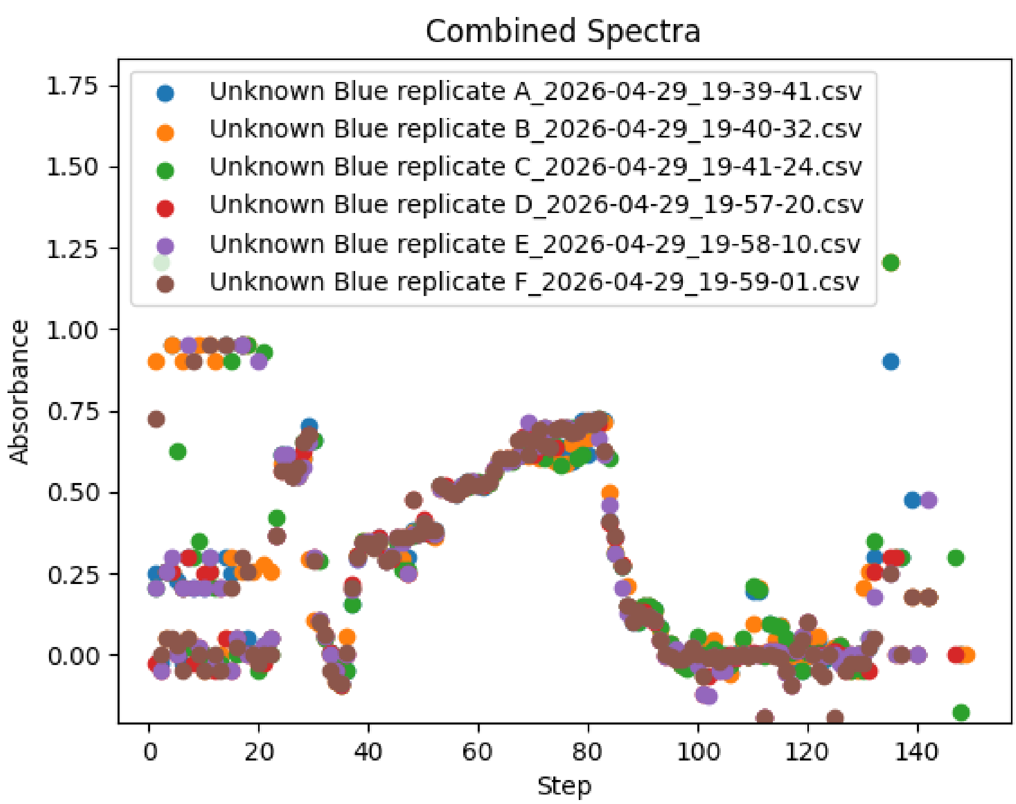
Standard Blue 100% replicate C_2026-04-29_19-26-47.csv
Absorbance
Step


Combined Spectra
Unknown Blue replicate A_2026-04-29_19-39-41.csv
Unknown Blue replicate B_2026-04-29_19-40-32.csv
Unknown Blue replicate C_2026-04-29_19-41-24.csv
Unknown Blue replicate D_2026-04-29_19-57-20.csv
Unknown Blue replicate E_2026-04-29_19-58-10.csv
Unknown Blue replicate F_2026-04-29_19-59-01.csv
Absorbance
Step

2.2 – Spectra (Serial dilutions)

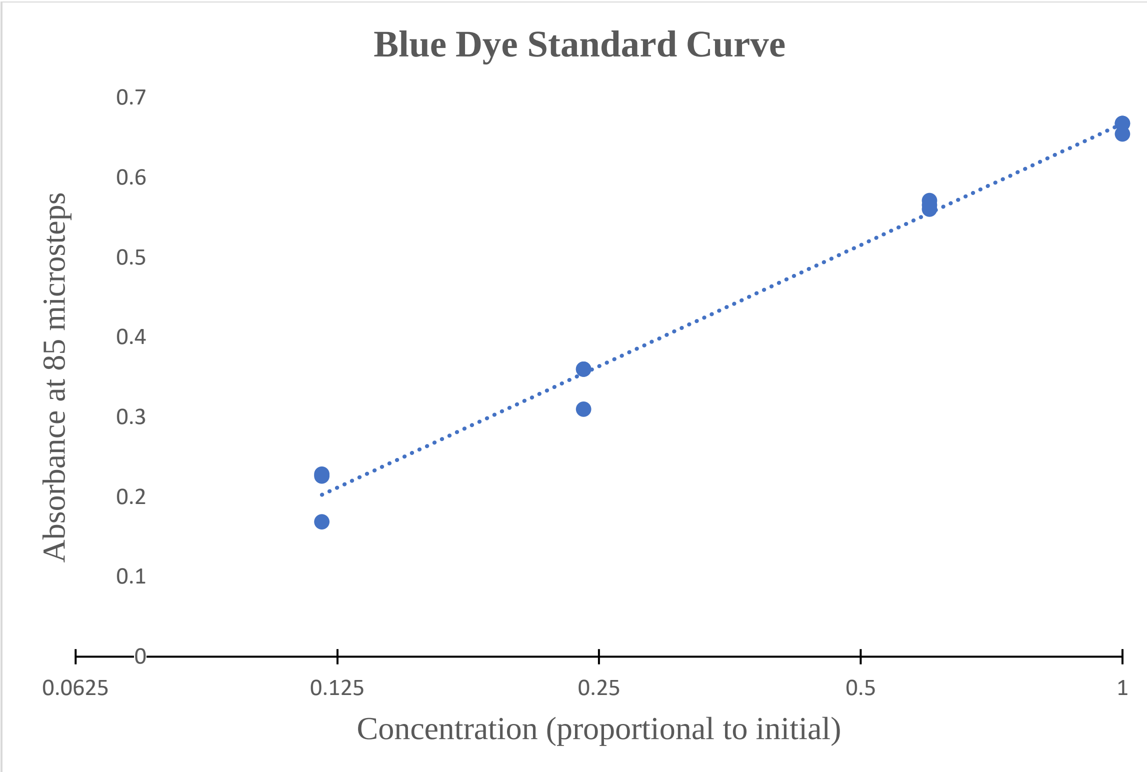
The replicate spectra of the diluted blue dye (previous page) were again similar enough within themselves to be adjudged successfully identified by the spectrophotometer. Unfortunately, comparing different dilutions reveals very little change in absorbance values. All four dilutions flatten at a fairly constant absorbance between .7 and .8 in a stretch from indices 60 to 80 or thereabouts. This suggests that the dilutions are too strong, and that observable absorbance reaches its maximum far before the actual λ_{max}. The one peak that can clearly be observed to get higher is the one around index 85, which is very low in the 12% dilution, but gradually gets higher with increasing concentration. Because the peak is legitimately at about the highest limit of quantitation in the 100% dilution, this makes it perfect to analyze.

2.3 – Standard curve and identification of the “unknown”

When this peak was quantitatively examined, however, it was seen that although higher concentrations of blue dye generally corresponded to higher absorbance, this was at no time a straightforward linear relationship as is anticipated by Beer’s Law

$$A = \varepsilon b C$$

This is likely due to the initial concentration of dye being far past the range of linear response (it had deliberately been set high to proactively mitigate the spectrophotometer’s low precision). Since the data at index 85 appeared strongly to display a steady flattening, this point was used as a reference point for analysis.



A logarithmic trendline was proposed to account for the limited response of absorbance to increasing concentration after the endpoint of linearity (the graph above has a logarithmically scaled x-axis, disguising the logarithmicity of the trendline). This trendline had an R^2 value of .9873, and the equation

$$A = 0.2192 \ln(C) + 0.6676$$

where A is absorbance in arbitrary units and C is proportional concentration. This formula was used on the measured absorbance of the six replicate trials of the unknown at index 85.

sample	A	B	C	D	E	F
absorbance	0.56040	0.31585	0.30986	0.36701	0.30986	0.36027
concentration (calculated)	0.61321	0.20095	0.19554	0.25377	0.19554	0.24609
	With outlier			Without outlier		
mean:		0.28418			0.21838	
std dev:		0.16327			0.02902	
range:	0.12092	to	0.44745	0.18936	to	0.24740

A one-tailed Grubb's test reveals that the concentration calculated based on the absorbance of unknown replicate A was statistically significant as an outlier:

$$G_{test} = \frac{|X - \bar{x}|}{s} = \frac{|.61321 - .28418|}{.16327} = 2.016$$

$$G_{crit}(\alpha = .01; \mathrm{n} = 6) = 1.944$$

Since $G_{test} > G_{crit}$, A was not accounted for in calculating mean and standard deviation, giving 21.8 % ± 2.9 %, or a range of 18.9 % to 24.7 %. This result was obtained without referencing the identity of the unknown (so that it remained an unknown, making this technically a double-blind study). Reference to the sheet containing the recipe for the unknown revealed that it was made by diluting 8 mL of 24% solution to 10 mL, creating a final concentration of

$$\frac{24\% \cdot 8\ \mathrm{mL}}{10\ \mathrm{mL}} = 19.2\%$$

Since 19.2% is within the predicted range (mean ± standard deviation), the test was deemed successful.

3 – Discussion: Unexplored Avenues

Archimedes famously said, "Give me a lever long enough, and a place to stand, and I shall move the world." Well, given a semester long enough, and a lab full of old parts the entire field of spectroscopy could have been reengineered from first principles. Unfortunately, the semester was not longer than it was, and so a visible spectrophotometer with partial function was all that was accomplished. Had slightly more time been available, however, the range of function could significantly have been increased, without requiring many more supplies. Here, then, are some directions for future exploration.

3.1 – Manual Calibration

The most obvious way that results could be improved is to calibrate this instrument with a professional spectrophotometer so that actual wavelength values (even if approximate) might be assigned to motor positions, removing the arbitrary ‘indices’ on all of the graphs. Theoretically this would not even be particularly difficult: simply record the spectrum of a molecule with two distinct peaks (possibly Bromothymol Blue dye), using both spectrophotometers. Then correlate the wavelength of the first peak from the professional instrument to the index of that peak from the janky one, and likewise with the second. Given two reference points, everything else is guaranteed to follow. The one caveat is that the solution would have to be dilute enough that the janky instrument can actually detect its λ_{max}es, but that should not be difficult. This calibration could be easily tested as well by comparing the calculated value for another peak to the professional instrument’s number.

3.2 – Automatic Calibration

The only problem with the above is that when the driver is not supplying power to the motor (i.e. when the RPi is not running the spectrophotometer script), the motor shaft can easily be bumped or moved, rendering any calibration transient and dooming one to an eternity of recalibration. However, if the appropriate solution (e.g. Bromothymol Blue in a buffer solution) were kept on hand in moderate quantities, a Calibrate command could fairly easily be added to the Blank and Sample already available. After scanning the reference sample, the index values of the two peaks (of known wavelength) could be automatically detected (as the two local maxima) and used to determine how much (if at all) the motor had been rotated since its last calibration. Then it would be a simple matter of stepping forward or backward the exact right number of microsteps to bring it back into alignment.

3.3 – Wavelength Monitoring

With a spectrophotometer capable of precise wavelength adjustment, the next step would obviously be to create a program capable of turning to a precise wavelength and measuring absorbance at that particular wavelength. This would enable quantitative analysis without having to sift through long spectral files; it could possibly even be controlled through the RPi’s command line, since it is only outputting a signal number (absorbance at wavelength X) per sample.

Acknowledgements

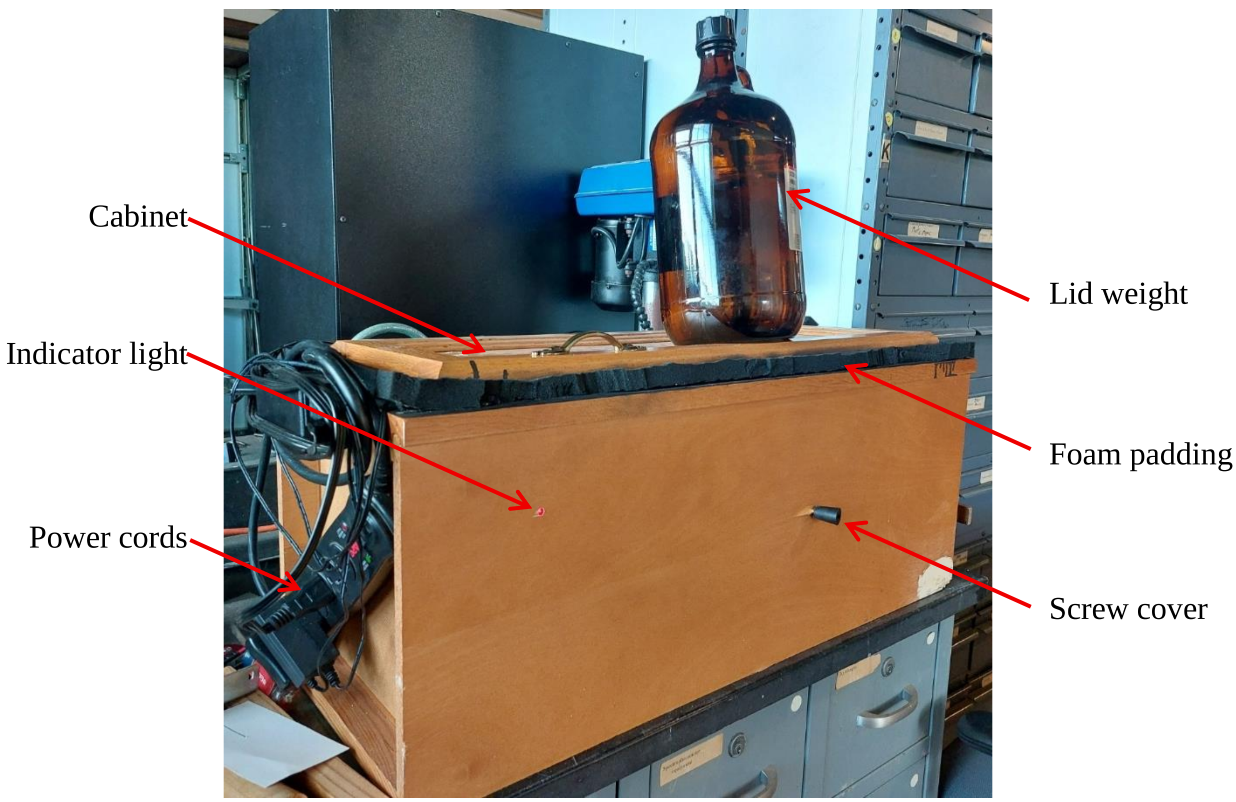
This project could not have been completed (at least not by me) without help. Accordingly, I would like to take this opportunity to thank

- **Dr. WS** for giving me freedom to use anything in his lab in the name of science, for listening to me vent every time something didn't work and exult every time something did, and for constantly inspiring to dream bigger and think outside the box;
- **Dr. LW** for sharpening my thinking and refining my plans by asking all the right questions, for holding me accountable to a limited timeline when all I really wanted to do was procrastinate, and for tempering Dr. Shafer's enthusiasm with a touch of realism;
- **SB** for patiently answering dumb questions like, "what is a Raspberry Pi?" and for inventing a complete workflow for the electronics one night we both were working late in the lab;
- **LC** for creating and analyzing the standards and "unknown" used in the final test, and for helping troubleshoot when the spectrophotometer malfunctioned;
- **SS** for trying valiantly to recover a stolen package of parts, and for her three months of patience working in closer proximity to me than anyone else had to;
- **All of the above** for their polite interest, useful feedback, and uplifting encouragement throughout the semester; and above all,
- **God**, to Whose help I entirely attribute the project coming together in the final two weeks after three months of delays and setbacks.

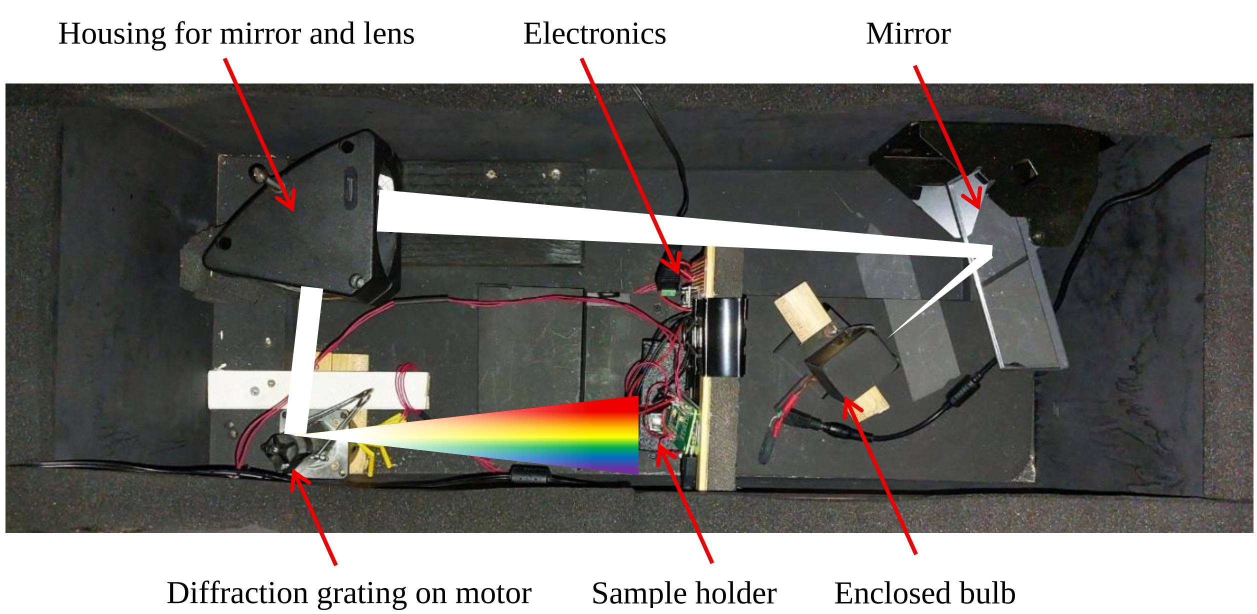
Supporting Information

S.1 – Images

S.1.1 – Exterior



S.1.2 – Interior



S.2 – Complete Parts List

Part	Use	Origin
12 V @ 1 A power adapter	Motor power	Repurposed (Wi-Fi router)
12 V @ 6 A power adapter	Bulb power	Purchased
5.1 V @ 1 A power adapter	RPi power	Already owned
Analog-Digital Converter	Data mediator	Purchased
Binder clips (assorted)	Stabilization	Already owned
Black enamel spray	Interior light / heat protection	Already owned
Black spray paint	Interior light protection	Purchased
Bolts	Attachment	Already owned
Bracket	Cord management	Repurposed (overhead projector)
Breadboards	Parts mounting	Already owned / Purchased
Capacitor (electrolytic)	Photodiode power regulator	Repurposed (unknown circuit board)
Capacitors (ceramic)	Photodiode power regulator and low-pass filter	Already owned
Cuvette holder	Holder for cuvettes	Repurposed (spectrophotometer)
Diffraction grating	Light fracturing	Repurposed (spectrophotometer)
Foam sheets	Padding	Repurposed (unknown prior usage)
Foam strips	Light protection around lid	Repurposed (case insert)
Four-liter bottle of deionized water	Lid weight	Already owned
Gorilla glue	Attachment	Already owned

Heat shrink wire connectors	Electrical insulation	Already owned
Laptop / Desktop computers	Data visualizer	Already owned
Lens	Beam focus	Repurposed (overhead projector)
Metal box	Bulb capsule	Repurposed (spectrophotometer)
Mirrors	Beam diverter	Repurposed (overhead projectors)
Nail	Cord wrapping	Already owned
Nuts	Attachment / Stopgap	Already owned
Photodiode w/ Transimpedance Amplifier	Light sensor	Purchased
Power strip	Switchable master power	Already owned
Raspberry Pi Zero W	Driver controller, data acquirer and conveyor	Purchased
Red LED	Readiness indicator	Repurposed (unknown circuit board)
Resistor	LED protector	Repurposed (unknown circuit board)
Rubber stopper	Personal protection	Already owned
Ruler fragments	Motor shims and base for lamp	Repurposed (rulers)
Screws	Attachment	Already owned
Slit	Beam narrower	Repurposed (spectrophotometer)
Small cabinet	Box for spectrophotometer	Repurposed (biology lab)
Stepper driver	Motor controller	Purchased
Stepper motor	Spectrum control	Repurposed (unknown prior usage)
Thermistor	Thermal resistor	Purchased

Tungsten-Halogen Bulb	Light source	Repurposed (spectrophotometer)
Wire	Electrical connection	Already owned
Wood blocks/planks	Part alignment	Repurposed (scrap wood)
Wooden box	Photodiode capsule	Repurposed (radioactive sample storage)

S.3 – Instructions for Use[10]

1. Switch on the power strip on the side of the cabinet
2. While the bulb warms up and the RPi boots, remove the sample holder box from its position underneath the RPi and in front of the photodiode
3. Put a cuvette of deionized water into the sample holder and replace it precisely in the cabinet; the box's unattached hinge flap should be touching the inside front wall of the cabinet, and the box's open back should be covering the photodiode
4. Once the red light in the front of the cabinet has turned on, open a terminal on the controlling computer and type

   `ssh spectrophotometer@strawberry-pie`
5. If prompted for a password, use `Ea-Nasir`
6. When the RPi's command line input appears, type

   `source adc-env/bin/activate`
7. Next, type

   `python spectrophotometer.py`

If you have done everything right so far, you will see something like this:

[10] These instructions assume access to a computer already set up to interact with the spectrophotometer. This means

- The controlling computer must be able to ssh (Secure Shell) into the RPi's terminal
- The RPi must have permission to write files into the computer's directory by SCP without needing a password
- The computer must be able to run the `spectrophotometer_control.py` (and preferably also `spectrum_compare.py`) script

```
PS C:\Users\[redacted]> ssh spectrophotometer@strawberry-pie
Linux strawberry-pie 6.12.47+rpt-rpi-v6 #1 Raspbian 1:6.12.47-1+rpt1 (2025-09-16) armv6l

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Apr 29 23:40:46 2026 from [redacted]
spectrophotometer@strawberry-pie:~ $ source adc-env/bin/activate
(adc-env) spectrophotometer@strawberry-pie:~ $ python spectrophotometer.py
standby...
```

The “standby” means that the spectrophotometer is now waiting for a cue from your computer.

8. Open a second terminal window on the computer
9. Navigate to the directory where the `spectrophotometer_control.py` script is, using the `cd` command[11]
10. Run the script using `python spectrophotometer_control.py`
11. A window with two buttons and a textbox will open on the computer screen; if the spectrophotometer has not been blanked recently, or if the RPi just rebooted, make sure the sample holder contains a deionized water blank, close the cabinet lid, place a heavy object[12] on top to hold the lid tightly shut, and click “Blank”
12. The RPi’s terminal will display a crude spectrum scrolling upwards as the stepper turns the diffraction grating and the photodiode logs its baseline data
13. When this stops, replace the blank with your sample, replace the sample holder in the cabinet, close the lid, etc.

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[11] For example, if you are currently in `C:\Users\Lab`, and the full filepath is `C:\Users\Lab\Instruments\Janky\spectrophotometer_control.py`, you can get to the appropriate directory by typing `cd ”C:\Users\Lab\Instruments\Janky”`

[12] e.g. a four-liter bottle of deionized water, or whatever industrial-strength chemical happens to be lying around

14. Type a clearly identifiable name into the textbox on the computer screen[13] and hit Sample
15. Like with the blank, the RPi’s terminal will again display a rough spectrum;[14] unlike with the blank, at the end of this one, it will send a full `.csv` file with its readings to the computer and when the `spectrophotometer_control.py` script sees this appear, the absorbance spectrum will automatically appear on the screen
16. Repeat steps 13-15 at will, remembering to update sample names/descriptions, and recognizing that the new spectrum will only automatically appear after you have closed the last one
17. If you should ever desire to look at an old spectrum again, or to compare more than one spectrum, that is where `spectrum_compare.py` comes in; open a new terminal, navigate to the correct directory and execute the script in the same way as with `spectrophotometer_control.py`
18. This time the window will only have one button: “Open and Plot Spectra”; click it and select the samples you wish to compare; the absorbance data will be plotted on one chart
19. You can rescale the graph to explore it more closely using the buttons at the bottom of the window; you can also open multiple graphs at once as long as you click the “Open and Plot Spectra” button again
20. All the Python scripts used in this project are very simple, so feel no trepidation about editing them to adapt the functionality of the instrument; the sky is very nearly the limit!

[13] things like, “Edmond Dantes red unknown .5 M dilution replicate A” work well; things like “Sample” less so; “blank” will just make it run another blank

[14] n.b.: this is not an absorbance spectrum, but merely a light intensity monitor, and is therefore of little use except (hopefully) to forestall boredom